

Chapter 3

Data Modeling Using the Entity-Relationship (ER) Model

Exam Preparation Notes

Fundamentals of Database Systems | Elmasri & Navathe, 7th Edition

Note: This chapter carries relatively lower exam weightage — focus on key concepts, notation, and examples.

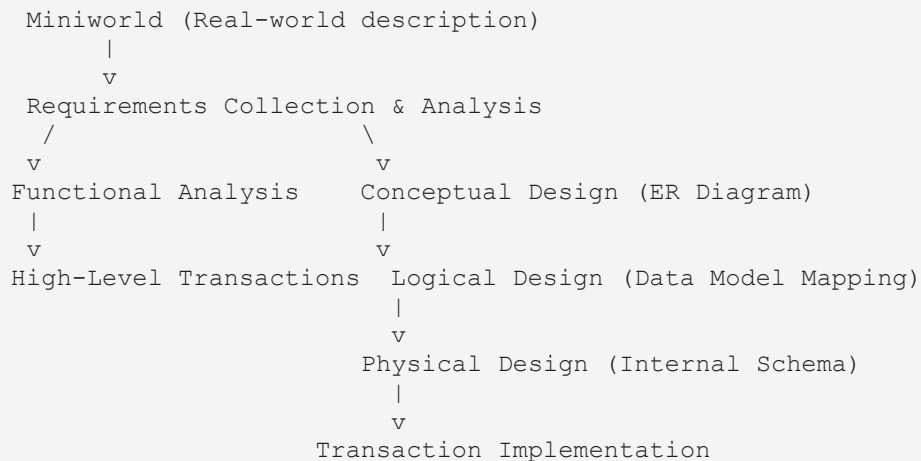
1. Overview of Database Design

Two main activities in database design:

- Database Design – focuses on the structure of data (this chapter)
- Applications Design – focuses on programs/interfaces that access the database (software engineering)

1.1 Phases of Database Design

The design process moves from a real-world description (miniworld) through several stages:



- Conceptual Design produces an ER Diagram (schema) that is DBMS-independent
- Logical Design maps the ER schema to a DBMS-specific model (e.g., relational tables)
- Physical Design optimises for storage and performance

1.2 Three-Schema Architecture (Recap)

Level	Schema	Data Model
External	External Views (user views)	Representational Data Model
Conceptual	Conceptual Schema (ER / relational)	Conceptual Data Model
Internal	Internal Schema (physical storage)	Physical Data Model

2. ER Model Core Concepts

The ER model has three fundamental building blocks:

- Entities
- Attributes
- Relationships

2.1 Entities

Entity: A specific, distinguishable object or thing in the real world that is represented in the database.

Examples:

- Employee John Smith
- The Research Department
- Project ProductX

Entity Type: A category/schema definition grouping entities with the same structure and attributes.

- Example: EMPLOYEE, DEPARTMENT, PROJECT are entity types

Entity Set: The current collection of all entity instances stored in the database for a given entity type (the current state).

□ **Exam Tip:** Know the distinction: Entity Type = schema definition; Entity Set = actual stored data at any point in time.

2.2 Attributes

Attribute: A property that describes an entity. Every entity has a value for each of its attributes.

Example – EMPLOYEE attributes: Name, SSN, Address, Sex, BirthDate

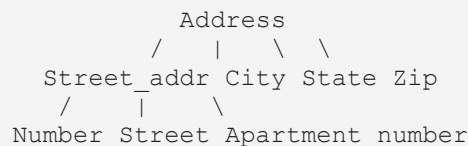
Types of Attributes

Type	Description	Example
Simple (Atomic)	Single indivisible value per entity	SSN, Sex, Salary
Composite	Made up of sub-components that can themselves be attributes	Name (Fname, Minit, Lname); Address (Street, City, State, Zip)
Multi-valued	Entity may have multiple values; denoted { }	{ Color } of a CAR; { PreviousDegrees } of a STUDENT
Derived	Value computed from other attributes; shown with dashed oval	Age (derived from BirthDate); Number_of_employees
Key	Uniquely identifies each entity; underlined in ER diagram	SSN for EMPLOYEE; VehicleID for CAR

Composite and multi-valued attributes can be nested:

- { PreviousDegrees (College, Year, Degree, Field) } — composite multi-valued attribute

Composite Attribute – Diagram



□ **Exam Tip:** A Key Attribute is underlined in ER diagrams. An entity type can have more than one key (e.g., CAR has Vehicle_id and Registration).

2.3 Value Sets (Domains)

Every simple attribute is associated with a value set (domain) — the set of allowed values.

- Analogous to data types in programming languages (integer, string, date, enum, etc.)
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3. ER Diagram Notation

ER diagrams are the standard graphical tool for representing conceptual schemas.

Symbol / Shape	Meaning
Rectangle	Entity Type
Double Rectangle	Weak Entity Type
Diamond	Relationship Type
Double Diamond	Identifying Relationship (for weak entity)
Oval	Attribute
Oval with underlined text	Key Attribute
Double Oval	Multi-valued Attribute
Dashed Oval	Derived Attribute
Oval connected to multiple ovals	Composite Attribute
Double line between entity & relationship	Total Participation (Existence Dependency)
Single line	Partial Participation
1, N, or M on line	Cardinality ratio
(min, max) near entity	Structural constraint on participation

ER Diagram – EMPLOYEE Entity Type (Example)



4. Relationships and Relationship Types

4.1 Basic Concepts

Relationship: An association between two or more distinct entities with a specific meaning.

- Example: Employee John Smith WORKS_ON Project ProductX
- Example: Employee Franklin Wong MANAGES the Research Department

Relationship Type: Schema definition of a relationship — identifies the name and participating entity types.

Relationship Set: All current relationship instances in the database (the current state).

Degree of a relationship: Number of participating entity types.

- Degree 2 = Binary (most common)
- Degree 3 = Ternary
- Degree n = n-ary

□ **Exam Tip:** In ER diagrams, a relationship type is shown as a diamond-shaped box connected to participating entity types by straight lines.

4.2 Recursive (Self-referencing) Relationships

A relationship where the same entity type participates more than once in different roles.

- Example: SUPERVISION — EMPLOYEE supervises another EMPLOYEE
- Role labels are shown on the connecting lines (e.g., 'Supervisor' and 'Supervisee')

```
      Supervisor          Supervisee
      1 /                  \ N
[EMPLOYEE] -----<SUPERVISION>----- [EMPLOYEE]
```

4.3 Attributes of Relationship Types

A relationship type can have its own attributes.

- Example: WORKS_ON has an attribute Hours (hours per week an employee works on a project)
- The value of Hours depends on a specific (Employee, Project) pair — it belongs to the relationship, not either entity alone
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5. Constraints on Relationships

5.1 Cardinality Ratio (Maximum Participation)

Specifies the maximum number of relationship instances an entity can participate in.

Cardinality	Meaning	Example
1:1	One entity in A relates to at most one in B, and vice versa	EMPLOYEE MANAGES DEPARTMENT
1:N	One entity in A relates to many in B; each in B relates to at most one in A	DEPARTMENT has many EMPLOYEEs (WORKS_FOR)
M:N	Many in A relate to many in B	EMPLOYEE WORKS_ON PROJECT

Visual Representations

```
N:1 (Many-to-one)           M:N (Many-to-many)

[EMPLOYEE]--N--<WORKS_FOR>--1-- [DEPARTMENT]

[EMPLOYEE]--M--<WORKS_ON>--N-- [PROJECT]
```

5.2 Participation Constraint (Minimum Participation)

Specifies whether every entity must participate in at least one relationship instance.

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Example: Every EMPLOYEE must work for a DEPARTMENT (total participation). A DEPARTMENT may or may not have a manager (partial participation for EMPLOYEE in MANAGES).

5.3 The (min, max) Notation

Specifies: Each entity participates in at least min and at most max relationship instances.

- Default: min = 0, max = n (no constraint)
- Must satisfy: $\min \leq \max$, $\min \geq 0$, $\max \geq 1$
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```
Example 1 - MANAGES:
[EMPLOYEE]-(0,1)-<MANAGES>-(1,1)-[DEPARTMENT]
  An employee manages at most 1 dept (0,1)
  A department must have exactly 1 manager (1,1)
```

```
Example 2 - WORKS_FOR:
[EMPLOYEE]-(1,1)-<WORKS_FOR>-(1,N)-[DEPARTMENT]
  An employee works for exactly 1 dept (1,1)
  A department has 1 or more employees (1,N)
```

□ **Exam Tip:** (0, 1) = optional, at most one. (1, 1) = mandatory, exactly one. (0, N) = optional, any number. (1, N) = mandatory, at least one.

6. Weak Entity Types

Weak Entity Type: An entity type that does not have its own key attribute; it depends on another (owner/identifying) entity for identification.

6.1 Key Characteristics

- Does not have a key attribute of its own

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- Total participation of the weak entity in the identifying relationship (always double line)

6.2 Identification

A weak entity instance is uniquely identified by the combination of:

- Its partial key value
- The specific owner entity it is related to

6.3 Example

DEPENDENT is a weak entity type:

- Depends on EMPLOYEE (the identifying entity type)
- Identified via the DEPENDENT_OF relationship (identifying relationship)
- Partial key: Name of the DEPENDENT
- To uniquely identify: Employee's SSN + Dependent's Name

```
[EMPLOYEE] ===<DEPENDENT_OF>=== [[DEPENDENT]]
      1                               N
      (Ssn)                        (Name) (Sex) (Bdate) (Relationship)
                                   ~~~~~ <- partial key (dashed underline)
```

□ **Exam Tip:** Weak entity = double rectangle. Identifying relationship = double diamond.
Partial key = dashed underline in oval.

7. COMPANY Database – Case Study

7.1 Requirements Summary

A company database needs to track:

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7.2 Entity Types Identified

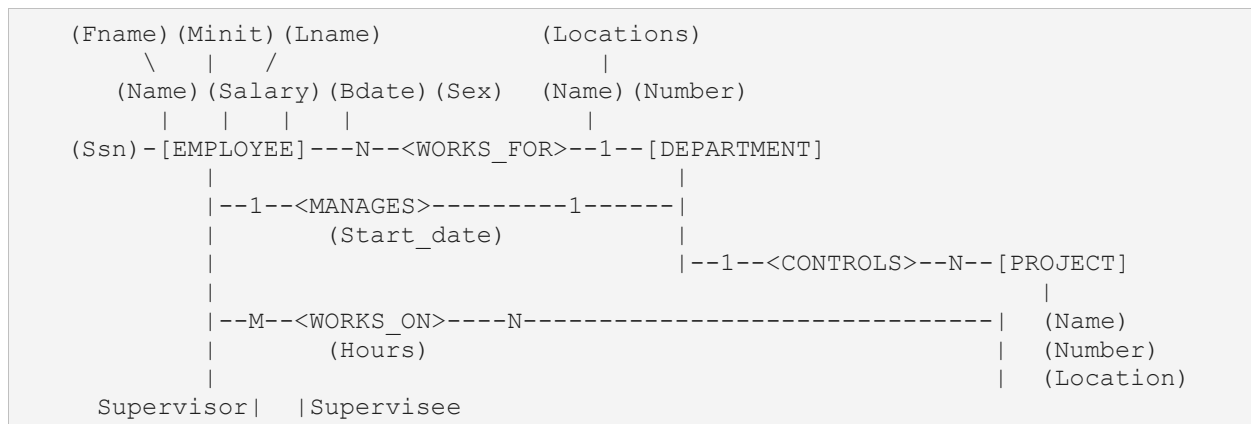
- DEPARTMENT
- PROJECT
- EMPLOYEE

- DEPENDENT (weak entity)

7.3 Relationship Types Identified

Relationship	Participating Entities	Cardinality	Notes
WORKS_FOR	EMPLOYEE, DEPARTMENT	N:1	Each employee works for exactly one dept
MANAGES	EMPLOYEE, DEPARTMENT	1:1	Each dept has one manager; employee may manage 0 or 1
CONTROLS	DEPARTMENT, PROJECT	1:N	Each project controlled by one department
WORKS_ON	EMPLOYEE, PROJECT	M:N	Attribute: Hours. Employee works on many projects
SUPERVISION	EMPLOYEE (recursive)	1:N	One supervisor, many subordinates
DEPENDENT_OF	EMPLOYEE, DEPENDENT	1:N	Identifying relationship for weak entity DEPENDENT

7.4 ER Diagram – COMPANY (Textual Representation)




```
1--<SUPERVISION>--N
|
|--1--<DEPENDENT_OF>--N--[[DEPENDENT]]
                        (Name) (Sex) (Bdate) (Relationship)
```

8. Relationships of Higher Degree

8.1 Binary vs Ternary

Binary relationship: Degree 2 (two participating entity types) — most common.

Ternary relationship: Degree 3 (three participating entity types).

□ **Exam Tip:** An n-ary relationship is NOT equivalent to n binary relationships. They can represent different information.

8.2 Example: SUPPLY (Ternary)

A SUPPLIER supplies a PART to a PROJECT — this three-way association cannot be split into two binary relationships without losing information.

```
[SUPPLIER] --<SUPPLY>-- [PROJECT]
|
[PART]
(Quantity) <- attribute of the ternary relationship
```

- 3 binary relationships (SUPPLIES, CAN_SUPPLY, USES) carry different and less combined information than the single SUPPLY ternary
- A ternary relationship can sometimes be modelled as a weak entity type with multiple identifying relationships

8.3 Constraint Notation for Higher-Degree Relationships

- Both (min, max) and 1/M/N labels may be needed to fully describe constraints
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- Overall, higher-degree constraints are harder to specify and can be ambiguous

9. Alternative Notations

9.1 UML Class Diagrams

UML Class Diagrams are widely used in industry for conceptual database design and OO software design.

Concept	ER Notation	UML Class Diagram
Entity Type	Rectangle	Class box (3 sections: name, attributes, operations)
Relationship	Diamond	Association line between classes
Cardinality	1, N, M on lines	Multiplicity: 1..1, 0..*, 1..* etc.
Weak Entity	Double rectangle	No direct equivalent; modelled differently
Attribute	Oval	Listed inside class box
Operations	Not in ER model	Listed in third section of class box

Key difference: UML class diagrams include operations (methods) — not present in the basic ER model.

9.2 (min, max) vs Cardinality Ratio Notation

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- Commercial tools often use crow's foot notation or UML multiplicity rather than classic ER shapes

10. Step-by-Step ER Design Methodology

Use this process when designing an ER schema from a textual requirements description:

Step	Action	COMPANY Example
1	Identify entity types from nouns in requirements	EMPLOYEE, DEPARTMENT, PROJECT, DEPENDENT
2	Identify attributes for each entity type	EMPLOYEE: SSN, Name, Address, Sex, BirthDate, Salary
3	Identify key attributes (uniquely identify each entity)	SSN for EMPLOYEE; Number for DEPARTMENT
4	Identify relationship types from verbs/associations	WORKS_FOR, MANAGES, WORKS_ON, CONTROLS
5	Determine cardinality ratios and participation constraints	WORKS_FOR is N:1; EMPLOYEE has total participation
6	Identify weak entity types	DEPENDENT depends on EMPLOYEE

Step	Action	COMPANY Example
7	Refine initial design — move attributes that are really relationships	Manager of DEPT becomes MANAGES; Works_on becomes WORKS_ON

11. Quick-Reference: Key Exam Points

□ **Exam Tip:** This chapter has lower exam weightage — prioritise understanding core notation, entity/attribute types, relationship constraints, and weak entities.

11.1 Must-Know Definitions

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11.2 ER Notation Summary (Quick Flash)

SHAPE	MEANING
Rectangle	Entity Type
Double Rectangle	Weak Entity Type
Diamond	Relationship Type
Double Diamond	Identifying Relationship
Oval	Attribute
Underlined Oval	Key Attribute
Dashed-underline	Partial Key (weak entity)
Double Oval	Multi-valued Attribute
Dashed Oval	Derived Attribute
Double Line	Total Participation
Single Line	Partial Participation
1 / N / M	Cardinality Ratio
(min, max)	Structural Constraint

11.3 Common Mistakes to Avoid

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11.4 COMPANY Database Cheat Sheet

Relationship	Ratio	(min,max) Employee	Notes
WORKS_FOR	N:1	(1,1) / (1,N)	Employee works for exactly 1 dept; dept has 1+ employees
MANAGES	1:1	(0,1) / (1,1)	Employee may manage 0 or 1 dept; dept has exactly 1 manager
CONTROLS	1:N	N/A	1 dept controls N projects; each project has 1 controlling dept
WORKS_ON	M:N	(1,N) Worker	Employee works on 1+ projects; has Hours attribute
SUPERVISION	1:N (recursive)	(0,N)/(0,1)	Employee has 0 or 1 supervisor; supervisor has 0+ subordinates
DEPENDENT_OF	1:N	(0,N)	Identifying relationship for weak entity DEPENDENT